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Harnessing the Genetic Diversity of the Colombian Central Collection to Dissect Pigmentation Genetics in Andigenum Landraces

No debería explicitar que es la CCC de "papa"?

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Abstract

The Colombian Central Collection of potatoes (CCC) is one of the most diverse germplasm resources in the country and a primary source of genetic variability for breeding. Most accessions belong to the Andigenum group, which exhibits remarkable pigmentation diversity with associated nutraceutical potential. However, the genetic architecture of color traits in this group remains poorly characterized. In this study, we analyzed historical phenotypic data to investigate the genetic basis of coloration in andigenum accessions from the CCC. Color traits across tuber skin, tuber flesh, flowers, stems, and sprouts were assessed using the Royal Horticultural Society Color Chart. Genome-Wide Association Studies and genomic prediction were conducted using high-density SNP markers. We identified 21 significant SNP–trait associations, with 12 overlapping or located near loci previously reported for pigmentation or tuber-related traits, while the remainder represent novel associations. Most primary color traits showed moderate to high heritability

($h^2 > 0.55$), and genomic prediction achieved predictive abilities of up to 0.84 for flower primary color and 0.71 for stem color. These findings provide new insights into the genetic control of color traits in Andigenum potatoes, identify valuable molecular markers for selection, and underscore the importance of conserving and utilizing traditional landraces for the development of nutritionally enhanced cultivars and the advancement of food security initiatives.

Resumen: La Colección Central Colombiana de papa (CCC) es uno de los recursos de germoplasma más diversos del país y constituye una fuente primaria de variabilidad genética para el mejoramiento. La mayoría de las accesiones pertenecen al grupo Andigenum, el cual exhibe una notable diversidad de pigmentación con potencial nutracéutico asociado. Sin embargo, la arquitectura genética de los rasgos de color en este grupo sigue siendo poco caracterizada. En este estudio analizamos datos fenotípicos históricos para investigar la base genética de la coloración en accesiones de razas locales Andigenum de la CCC. Los rasgos de color en piel y pulpa de tubérculo, flores, tallos y brotes fueron evaluados utilizando la Royal Horticultural Society Color Chart. Se llevaron a cabo estudios de asociación de genoma completo (GWAS) y predicción genómica empleando marcadores SNP de alta densidad. Identificamos 21 asociaciones significativas entre SNP y rasgos, de las cuales 12 coincidieron o se ubicaron cerca de loci previamente reportados para pigmentación o características relacionadas con el tubérculo, mientras que la otra mitad representó asociaciones novedosas. La mayoría de los rasgos de color primarios mostraron una heredabilidad moderada a alta ($h^2 > 0{,}55$), y la predicción genómica alcanzó habilidades predictivas de hasta 0,84 para el color primario de la flor y 0,71 para el color del tallo. Estos hallazgos aportan nuevos conocimientos sobre el control genético de los rasgos de color en papas Andigenum, identifican marcadores moleculares valiosos para la selección y destacan la importancia de conservar y utilizar los cultivares tradicionales en el desarrollo de variedades con propiedades nutricionales mejoradas y en el fortalecimiento de las iniciativas de seguridad alimentaria.

Dividir mejor esta palabra

Keywords: Andigenum group, potato landraces, color traits, GWAS, Genomic Prediction

1 Introduction

Potato (*Solanum tuberosum* L.) is the most important root and tuber crop globally, the third most available staple food, and ranks sixth in overall crop production worldwide (Food and Agriculture Organization of the United Nations, 2024). It plays a vital role in food security (Devaux et al., 2021).Cultivated potatoes originate from both Andean and Chilean landraces, with Andean landraces—primarily belonging to the Andigenum group—being the most widely cultivated. This group exhibits remarkable morphological and genetic diversity (Spooner et al., 2005). Extensive studies across the Andean region have reported variation in color and nutritional traits, including research from Peru (Bellumori et al., 2020), Ecuador (Balladares and Ramos, 2018), Bolivia (Larico et al., 2024), Chile (Behn et al., 2023), and Colombia (Berdugo-Cely et al., 2017, 2023; Garnica H et al., 2012). Although pigmented potatoes have been linked to potential health benefits (Bvenura et al., 2022; Lachman and Hamouz, 2005), most commercial varieties are white-fleshed. Notably, the Andigenum group encompasses a rich diversity of pigmented types, highlighting the importance of elucidating the genetic basis of color in this group.

The study of potato color has a long history, with early research dating back to studies on the inheritance of flower and tuber skin and flesh color (Salaman, 1911; Lunden, 1932; Black, 1933; Dodds and Long, 1955; van Eck et al., 1994). More recent investigations have focused on identifying genes involved in tuber skin and flesh pigmentation, in both diploid (Parra-Galindo et al., 2019) and tetraploid potatoes (Zhang et al., 2009; Pandey et al., 2022; Yusuf et al., 2024), as well as exploring the associated regulatory networks (Cho et al., 2016; Bonar et al., 2018). Over time, advances in both color measurement technologies and genetic tools have improved the resolution and accuracy with which these traits can be studied. Moreover, beyond its role as a morphological trait, color has been associated with nutraceutical properties (Dodds and Long, 1955; Parra-Galindo et al., 2019; Li et al., 2024; Cho et al., 2016), particularly due to its link with compounds such as anthocyanins, thereby motivating further integrative research.

Plant color measurement techniques have evolved significantly over time. Initially, assessments were based on human perception, later transitioning to color matching using predefined charts. Currently, more accurate and objective methods include spectrometric and photographic color analysis (Kasajima, 2019). In the case of potato, morphological descriptors—including those for color traits—were developed by the International Potato Center (CIP) (Huaman et al., 1977; Gómez, 2000). These descriptors define categories for both primary and secondary colors and have been widely adopted in the characterization of potato diversity (Behn et al., 2023; Berdugo-Cely et al., 2017; Larico et al., 2024). Additionally, the Royal Horticultural Society (RHS) introduced a color chart that expanded the number of available color categories from 10 to 884 (Royal Horticultural Society, 2015). This chart has been widely used in plant color characterization (Tang, 2014), is easy to apply under field conditions, and represents an important step toward standardized color description. Colorimeters have also been used to characterize potato color traits (Thornton et al., 2013), and more recently, the use of photographs combined with computer vision techniques has emerged as a more precise and scalable approach (Miller et al., 2023; Yusuf et al., 2024; Azizi et al., 2016; Ropelewska, 2021; Feldman et al., 2024). Nevertheless, to ensure consistency in color characterization, it is essential to establish relationships between different color descriptors (Voss, 1992; Ayala-Silva and Meerow, 2006).

On the other hand, genetic research in potato has advanced considerably due to the availability of cost-effective, high-throughput genotyping platforms and key genomic resources such as the *Solanum tuberosum* reference genome (Consortium, 2011; Pham et al., 2020). The development of SNP arrays for potato (Felcher et al., 2012; Vos et al., 2015), along with the implementation of genotyping-by-sequencing techniques (Sharma et al., 2024), has made it feasible to characterize potato collections globally and apply these markers in Genome-Wide Association Studies (GWAS) and genomic prediction. These tools have enabled in-depth analyses of genetic diversity and the architecture of important traits (Sharma et al., 2024; Pandey et al., 2021). However, a significant portion of potato's genetic diversity—especially that of the Andigenum group—remains largely unexplored (Ghislain and Douches, 2020). Expanding genomic resources for this group and studies is crucial to gain a deeper understanding of its distinct genetic characteristics.

Este nombre debería estar en el título.

The Colombian Central Collection (CCC) of Potatoes, managed by AGROSAVIA, was established in 1950. It consists of 2,504 accessions conserved through botanical seed, *in vitro* methods, and field collections. The field collection includes 1,255 genetically diverse accessions, primarily landraces from the *Solanum tuberosum* group Andigenum 67.17%, followed by the Phureja 21.83% and tuberosum 5.9% (Manrique-Carpintero et al., 2023; Berdugo-Cely et al., 2017). Each year, the clonal collection is regenerated in the field near Bogotá, Colombia, where the curatorial team also characterizes its morphological diversity, including color diversity. To assess color-related traits, such as tuber skin and flesh, flower, stem, and berry, human visual assessment has been used in combination with morphological descriptors (Gómez, 2000) and the RHS Color Chart (Royal Horticultural Society, 2015). Although the descriptors of Gómez (2000) have been applied in GWAS(Berdugo-Cely et al., 2017, 2023), the color data derived from the RHS scale has not yet been integrated into genetic analyzes or linked to other datasets on color or nutritional and nutraceutical traits. Berdugo-Cely et al. (2023) identified a strong correlation between polyphenol and ascorbic acid contents and antioxidant capacity. Tubers with darker colors—primarily purple in tetraploids and red in diploids—showed the highest concentrations of these nutraceutical traits when pigmentation was present in both skin and flesh.

This study draws on historical color trait data to explore the genetic basis of coloration across multiple organs of the potato plant in landraces from the underexplored Andigenum group, recognized for its wide color variability. The CCC, composed primarily of tetraploid Andigenum landraces, has demonstrated high genetic diversity (Berdugo-Cely et al., 2017) and represents a valuable reservoir of genes for traits such as color variation that may enhance nutritional value. Using GWAS and genomic prediction, we examined whether the genes identified in the Andigenum group overlap with those previously reported in potatoes, including loci linked to nutraceutical traits. We identified 21 SNPs associated with color traits: 12 overlapped with or were located near variants reported in earlier studies of color, nutritional, or tuber traits, while the remaining represent novel targets for exploration—highlighting that the genetic architecture of color in potatoes remains incompletely understood. These findings provide new insights into the regulation of color traits in Andigenum potatoes, identify molecular markers useful for selection, and emphasize the importance of conserving collections such as the CCC and utilizing traditional landraces to develop nutritionally enhanced cultivars and strengthen food security initiatives. Understanding tuber pigmentation further holds promise for improving the nutritional value of potato, the world's third most important staple crop.

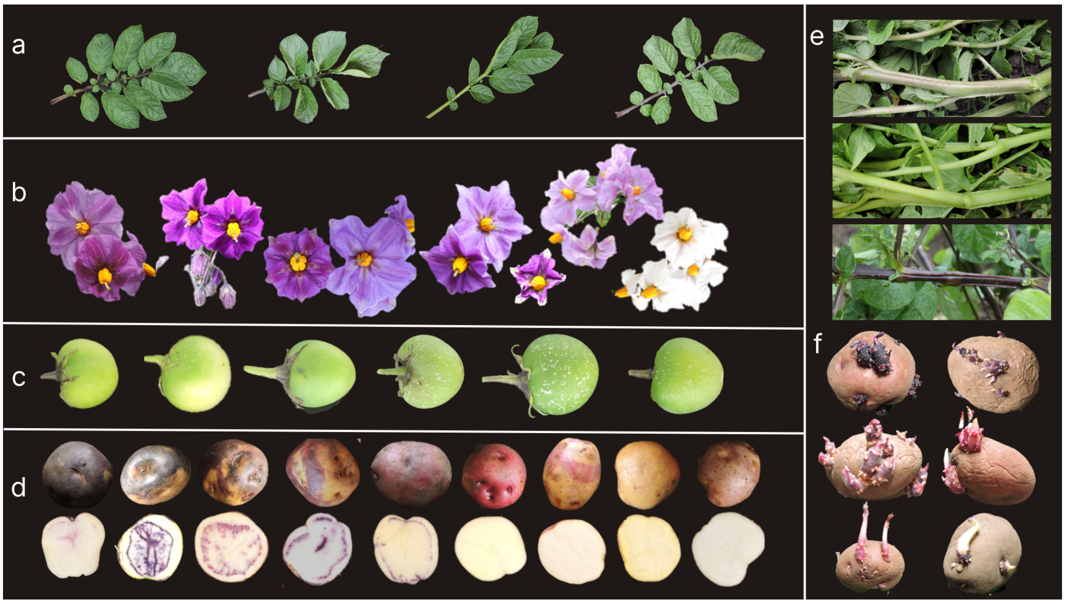

Fig. 1 Color diversity among CCC potato accessions observed in (a) leaves, (b) flowers, (c) berries, (d) tuber flesh and skin, (e) stems, and (f) sprouts.

2 Materials and Methods

The Colombian Central Collection of potatoes (CCC) represents one of the most diverse germplasm collections in the country and serves as the primary source of genetic variability for potato breeding in Colombia (Manrique-Carpintero et al., 2023). Currently, the CCC's clonal collection, maintained in the field, includes 1,255 accessions, of which 67.17% are landraces belonging to the *Solanum tuberosum* Andigenum group. Most of these accessions were collected prior to 1985. The CCC is regenerated annually in the municipality of Zipaquirá (see Figure 2), located in the department of Cundinamarca, Colombia, at an elevation of 2,950 meters, with an average temperature of 15°C and a relative humidity of 75% (Berdugo-Cely et al., 2017). The genotypic and phenotypic data used in this study are derived from accessions belonging to the Andigenum group within the CCC.

2.1 Genotypic data

A set of 599 tetraploid accessions from the CCC, belonging to the S. tuberosum Andigenum group, was previously genotyped (Berdugo-Cely et al., 2017) using the Potato Illumina Infinium SolCAP SNP array version I, which contains 8,303 SNPs. The array was processed on the Illumina HiScan SQ system (Illumina, San Diego, CA) at AGROSAVIA, and the ClusterCall R package (Schmitz Carley et al., 2017) was used to obtain the dosage genotype calls from polar coordinates (r, θ) obtained

Fig. 2 **UAV imagery of the CCC conserved in the field.** (a) Orthomosaic overview of the entire collection, generated from images captured by a P4 UAV at 12 m altitude in September 2022. (b) High-resolution view of individual plots at the flowering stage, where each furrow corresponds to one accession comprising 20 plants. North is oriented upward in both panels.

from GenomeStudio, applying default parameters, and calibration from F1 populations supported by the software. This resulted in a genotype call matrix (0: AAAA, 1: AAAB, 2: AABB, 3: ABBB, and 4: BBBB) for each SNP across all accessions.

Standard filtering criteria were applied to ensure data quality and reliability for subsequent GWAS analyses. SNPs with a minor allele frequency (MAF) below 1% were excluded to eliminate rare variants that may lead to spurious associations due to insufficient statistical power (Anderson et al., 2010). Individuals with a genotype call missing rate (MIND) greater than 10% were removed to avoid bias introduced by low-quality samples. SNPs with a missing rate (GENO) of $\geq 10\%$ were filtered to maintain marker integrity and reduce imputation noise. Additionally, SNPs with Hardy-Weinberg equilibrium (HWE) p-values below 1e-10 were excluded because extreme deviations may indicate genotyping errors, population structure artifacts, or selection (Wigginton et al., 2005). These thresholds are commonly used in polyploid GWAS studies and are consistent with established quality control practices (Lu et al., 2013; Pavan et al., 2020). After filtering, 4,641 SNPs were retained for downstream genomic analyses (Figure S1 shows the distribution of SNP density across the chromosomes used in the study, and File S1 provides the SNP marker matrix for the accessions analyzed).

2.2 Phenotypic data

The CCC is regenerated annually in furrows measuring 7 square meters, with 20 seed tubers planted per accession. Furrows are spaced 1 meter apart, and individual plants are separated by 0.35 meters (Figure 2). All accessions are managed under standardized agronomic practices based on soil analysis and weekly crop monitoring.

This regeneration process served as the basis for the morphological color characterization of 599 accessions. The evaluation involved assessing the uniformity of qualitative plant descriptors and examining the characteristics of randomly selected tubers at harvest within each furrow. Given the extensive size of the *Solanum tuberosum* Andigenum group collection and limited human resources, the 600 accessions were evaluated over a three-year period (2015–2017). Color traits were recorded for the stem (7 categories), berry (7 categories), primary and secondary flower colors (8 and 9 categories, respectively), tuber skin (9 and 10 categories, respectively), tuber flesh (8 and 9 categories, respectively), and sprouts (5 and 6 categories, respectively), using the descriptors proposed by Gómez (2000).

Additionally, in 2019, color characterization was performed using the fifth edition of the Royal Horticultural Society (RHS) color chart (Voss, 2002) to provide a more detailed color evaluation. This change expanded the codes from a maximum of ten options to a set of 884 codes. The chart was used to characterize the following nine traits: stem color (StemC), berry color (BerryC), primary flower color (PCFlower), primary tuber skin color (PCTuberskin), secondary tuber skin color (SCTuberskin), primary tuber flesh color (PCTuberflesh), secondary tuber flesh color (SCTuberflesh), primary sprout color (PCSprout), and secondary sprout color (SCSprout). The phenotypic RHS data for the accessions studied are provided in Supplementary File S2.

Moreover, the color characterization using the RHS color chart was converted to the Hue, Chroma, and Lightness (HCL or LCH) color space, as this model is closely aligned with human color perception and has been used in previous studies on potato color (Caraza-Harter and Endelman, 2020). The values of these LCH traits were derived from the transformation of the values of the potato color traits, measured using the Royal Horticultural Society (RHS) fifth edition color chart (http://rhscf.orgfree.com/) (Voss, 2002), into the three components: L, C, and H, of the Commission Internationale de l'Éclairage (CIE) LCH or CIE*L*C*h color space. Consequently, three components were obtained for each potato color trait, which will be referred to as LCH traits. Thus, the nine potato color traits were transformed into 27 LCH traits, which were used for genomic analyses

Correlation between two color descriptors

Two descriptors with different numbers of codes were used to characterize the color traits in CCC in different years. The correlation between the evaluations of both descriptors was calculated for each of the nine color traits using Cramer's V statistic (Lyman et al., 1986). Cramer's V measures the strength of the association between two qualitative variables, with values ranging from 0 to 1. Values below 0.6 indicate a weak association, while values of 0.6 or higher indicate a medium to strong association. To study the genetics underlying color, we selected traits with a medium-to-strong association, indicating genetic influence.

Linking Color to Nutritional and Nutraceutical Variables

This study also examined the association between color traits and nutritional and nutraceutical values using data from a previous study (Berdugo-Cely et al., 2023).

The nutritional traits of the potatoes were represented by the Ascorbic Acid Content (AAC), while the nutraceutical traits were: Total Phenol Content (TPC) and Antioxidant Activity (measured by 2.2-Diphenyl-1-picrylhydrazyl Radical Scavenging Activity - DPPH and Ferric Reducing Antioxidant Power - FRAP assays). The methodologies implemented to quantify these traits can be found in Berdugo-Cely et al. (2023). Given that the nutritional and nutraceutical variables were continuous, and the color traits were categorical, the determination coefficient was applied as a measure of association (Nagelkerke et al., 1991). To evaluate these associations, data from 282 Andigenum accessions in the CCC collection, which included information on both sets of traits, were analyzed.

2.3 GWAS analysis

A previous GWAS study by Berdugo-Cely et al. (2017) employed the same genetic dataset and color descriptors as those defined by Gómez (2000). In contrast, the present study used the RHS color chart as a phenotype for color characterization, increasing the number of trait values from 30 to 884, thus achieving greater resolution.

We utilized MultiGWAS (Garreta et al., 2021), a tool that integrates the outputs from multiple GWAS methods. Specifically, we incorporated the results from GWASpoly v1.3 (Rosyara et al., 2016) for tetraploid genotypes and GAPIT v3 (Wang and Zhang, 2021) for diploid genotypes. The analysis was conducted using the MultiGWAS Full Model (Q+K), which controls for population structure (Q, typically derived from principal component analysis) and kinship (K, based on a genetic relatedness matrix), thereby reducing the likelihood of false-positive associations.

Se usan sólo dos herramientas...pero más abajo habla de tres.

In MultiGWAS, each tool implements a mixed linear model (MLM) (Phenotype Genotype + Q + K), but with method-specific adjustments. GWASpoly adapts the MLM for polyploid data, incorporating population structure and kinship as fixed and random effects, respectively. GAPIT employs a standard MLM, treating the population structure (Q) as a fixed effect (e.g., from PCA) and kinship (K) as a random effect via its kinship matrix. By unifying these approaches, MultiGWAS robustly controls for population stratification and relatedness while detecting marker-trait associations across ploidy levels (diploid and tetraploid).

To address the issue of multiple testing, the MultiGWAS tool applies the Bonferroni correction. However, instead of adjusting the *p-values*, MultiGWAS adjusts the threshold to determine a significant *p-value*. Specifically, this threshold is set to α/m, where α represents the significance level and m is the number of tested markers from the genotype matrix. The CMPlot library (Yin et al., 2021) was used to generate Manhattan and Circos plots to visualize significant SNP markers.

Candidate Marker Identification

Given the evaluation of multiple traits and the integration of results from the four GWAS tools in MultiGWAS, we developed a proprietary scoring function called GSCORE. This function was designed to rank markers by incorporating the outputs from the MultiGWAS. Specifically, GSCORE selected the top 50 markers with the lowest p-values from each tool, resulting in 100 markers (2 tools × 50 markers). GSCORE is calculated as the sum of three weighted terms: inflation factor (I), contributing 70% of the score. Replicability (R) accounts for 10%, and significance (S) accounts for the remaining 20%. The scoring function is represented by the following equation:

$$GSCORE(M) = 0.7 * I + 0.1 * R + 0.2 * S$$

I is the inflation factor score, defined as $I = 1 - |1 - \lambda(M)|$, where $\lambda(M)$ is the inflation factor for marker M. This score is the highest when $\lambda(M)$ is close to 1. In addition, R is the number of SNPs shared among the three GWAS tools. Finally, S is a binary value (1 or 0) that indicates whether the SNP is significant (p-value i threshold). Marker M achieves a high $GSCORE$ when it has an inflation factor $\lambda(M)$ close to one, identifies a large number of shared SNPs across tools, and is statistically significant. In contrast, the score is low if $\lambda(M)$ is either too low (close to 0) or excessively high, identifies few shared SNPs, or is not significant. In other scenarios, the score is determined by a balance between the inflation factor, shared SNP count, and significance. This approach prioritizes markers that are statistically robust, consistent across tools, and highly significant.

Aquí habla de 3 herramientas pero arriba nombra sólo dos.

Error

Marker annotation

Markers were annotated by taking information from the Potato Genome Sequencing Consortium (PGSC) public data based on the double monoploid *S. tuberosum* Group Phureja from assembly DMv6.1(Pham et al., 2020) obtained from the SpudDB website. Annotation was performed by associating marker identifiers with information from both the high confidence gene models annotation, the InterProScan assigned GO terms for the working gene models, the InterProScan search results for the working gene models, and the SolCAP 69 K SNP positions on the DMv6.1 assembly available in SpudDB. Furthermore, for each trait, the amino acid sequences were analyzed using BlastKOALA (Kanehisa et al., 2016) for KEGG mapping to identify common functional categories among the significant markers.

....from both... esperaría dos elementos, pero aquí se nombran varios.

2.4 Heritability Estimation

Narrow-sense heritability (h^2), defined as the proportion of phenotypic variance explained by additive genetic effects (de los Campos et al., 2015), was estimated for each of the 21 LCH components derived from color traits. We used Bayesian Ridge Regression (BRR), implemented in the `BGLR` R package (Pérez and de los Campos, 2014), because the color-related traits analyzed in this study are expected to follow a polygenic architecture, with many loci of small effect contributing additively to the phenotype.

BRR assumes normally distributed genetic effects across markers and estimates heritability by repeatedly sampling the additive genetic variance (σ_a^2) and residual variance (σ_e^2) through a Bayesian process. The model was run for 15,000 iterations, with a burn-in of 5,000 and thinning every 5 steps. Heritability was computed at each iteration as:

$$h^2 = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_e^2}$$

This procedure yields a posterior distribution of heritability values for each trait, rather than a single point estimate. From this distribution, we report the posterior mean and the 95% credible interval. Unlike confidence intervals from frequentist methods, a Bayesian 95% credible interval indicates the range within which the true h^2 value lies with 95% probability, given the data and model.

2.5 Genomic Prediction (GP)

We applied Genomic Prediction (GP) to estimate the genetic potential of individuals for each of the 21 LCH color trait components. A total of seven statistical models were used, spanning parametric, semiparametric, and Bayesian approaches:

- **Parametric**: Genomic Best Linear Unbiased Prediction (GBLUP)
- **Semiparametric**: Reproducing Kernel Hilbert Spaces (RKHS)
- **Bayesian**: Bayesian Ridge Regression (BRR), Bayesian LASSO (BL), Bayes A, Bayes B, and Bayes C

All models were implemented using the `BWGS` R package (Charmet et al., 2020). Predictive performance was assessed via 5-fold cross-validation, repeated five times for each trait. Predictive ability was measured as the Pearson correlation between observed phenotypes and genomic estimated breeding values (GEBVs) across validation folds. All model parameters were set to the default values provided by the BWGS library.

2.5.1 GP using marker subsets

To determine the optimal number of markers for genomic prediction (GP) of each LCH color trait component, GP was performed using subsets containing between 1% and 100% of the markers. Markers were selected based on GWAS results and prioritized using our custom GSCORE scoring function (see Section 2.3). The aim was to assess whether predictive ability could be improved with a reduced marker set.

3 Results

All traits displayed distinct distribution patterns across the three components, with varying amounts of available data (Supplementary Figure S2). Component H showed the widest distribution for 8 of the 9 traits, whereas Components L and C exhibited similar ranges for 7 of the 9 traits and displayed bimodal distributions in 5 of them, including both tuber skin color traits. Berry color showed the narrowest distribution (0–120) across all three components, while the remaining eight traits spanned a broader range (0–350). Since no outliers were detected, all available data were retained for analysis.

Regarding data availability, 7 of the 9 traits had data for more than 90% of accessions. The two exceptions—secondary color of the tuber skin and tuber flesh—had fewer records because some accessions lacked these secondary colors. Nevertheless, even the trait with the least data had information from 94 accessions.

3.1 Correlation between descriptors

The correlations between color trait values measured using the two color descriptor systems (Figure 3A) showed that only berry color and primary tuber flesh color had weak correlations and were therefore excluded from the genetic analysis. In contrast, the remaining seven color traits exhibited moderate to strong correlations, with values ranging from 0.65 to 0.97. While some traits were correlated with each other, most of the moderate to strong correlations, as expected, were concentrated along the diagonal of the correlation matrix. Notably, strong correlations were observed between primary and secondary tuber skin colors, between primary tuber skin color and both primary and secondary tuber flesh colors, and between primary and secondary sprout colors.

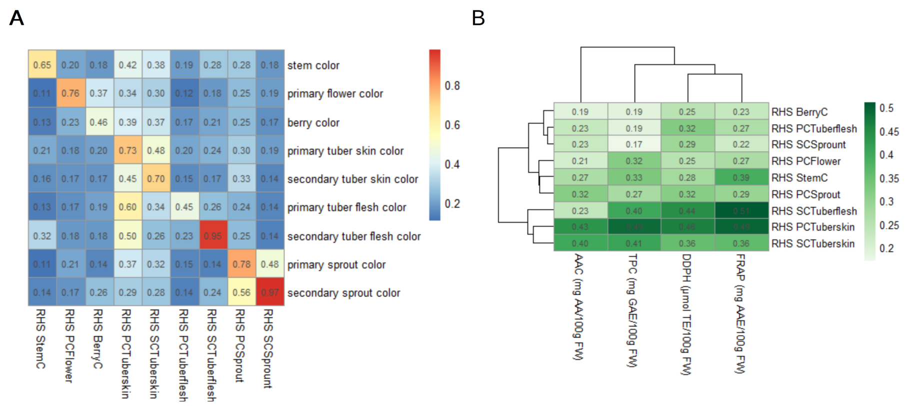

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Fig. 3 Correlation and determination coefficients of RHS potato color traits in CCC accessions. A. Heatmap showing the correlation matrix of potato color traits in CCC accessions, derived form descriptors by Gómez (2000) and measured with RHS (Voss, 2002) color chart. The color bar is located in the top right, with blue indicating low correlation values and red representing high correlation values. **B.** Heatmap showing the determination coefficient between nutritional (AAC) and nutraceutical (TPC, DPPH, and FRAP assay reported by (Berdugo-Cely et al., 2023) components and potato RHS color traits. The color bar is located at the top right, with the intensity of green representing the determination coefficient.

Parentesis sin cerrar.

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3.1.1 Linking Color to Nutritional and Nutraceutical Variables

In Figure 3B, the heatmap presents the determination coefficient R^2 between the color traits measured using the RHS color scale and nutritional/nutraceutical variables. R^2 below 0.3 reflect a weak association (blue cells), while R^2 between 0.3 and 0.51 reflects a moderate association between color traits and nutritional variables (yellow and red). Three specific traits—the secondary color of the tuber flesh and the primary and secondary colors of the tuber skin—exhibited notable associations with

Si es acerca de la Fig. 3B (Colores verdosos), entonces la descripción está erronea pq estos colores no los veo...blue cells...yellow and red....

nutritional variables. Antioxidant activity (measured using FRAP and DPPH assays) showed the strongest correlation with these traits. Additionally, Total Phenol Content and Ascorbic Acid Content demonstrated the highest association with the primary color of the tuber skin. Stem color, meanwhile, showed a moderate association with antioxidant activity and Total Phenol Content.

3.2 GWAS Analysis

GWAS analysis identified 21 significant SNPs associated with five traits: ten related to the secondary color of the tuber flesh, four to the primary color of the flower, three to the primary color of the sprout, two to the secondary color of the sprout, and two to the stem color (Table 1 and Figure 4). The Hue component and the secondary color of the tuber flesh showed the most significant SNP associations, which were also supported by previous reports (Pandey et al., 2022; Riveros-Loaiza et al., 2022; Sharma et al., 2024; Parra-Galindo et al., 2019). Of the 21 SNPs identified, one was previously reported as direct associations with potato color traits, while 11 were located within 1 Mb of loci (less than LD reported based on GBS data (Sharma et al., 2024; Berdugo-Cely et al., 2017)) previously reported for color, nutritional, or other tuber-related traits (Table 1 and Supplementary Table 1). In other words, 57% of the SNPs reported are consistent with the literature, while the other half should be studied.

The 21 associated SNP markers are annotated with 21 genes, represented by a total of 33 transcripts. A total of eight functional pathway categories were identified based on genes annotated with significant SNPs: seven belonged to the metabolism category and one to the genetic information processing category. Information on the primary annotations linked to these markers, as retrieved from the SPUD database, is provided in Supplementary Table 1, and the specific functional categories per trait are provided in Supplementary Table 2.

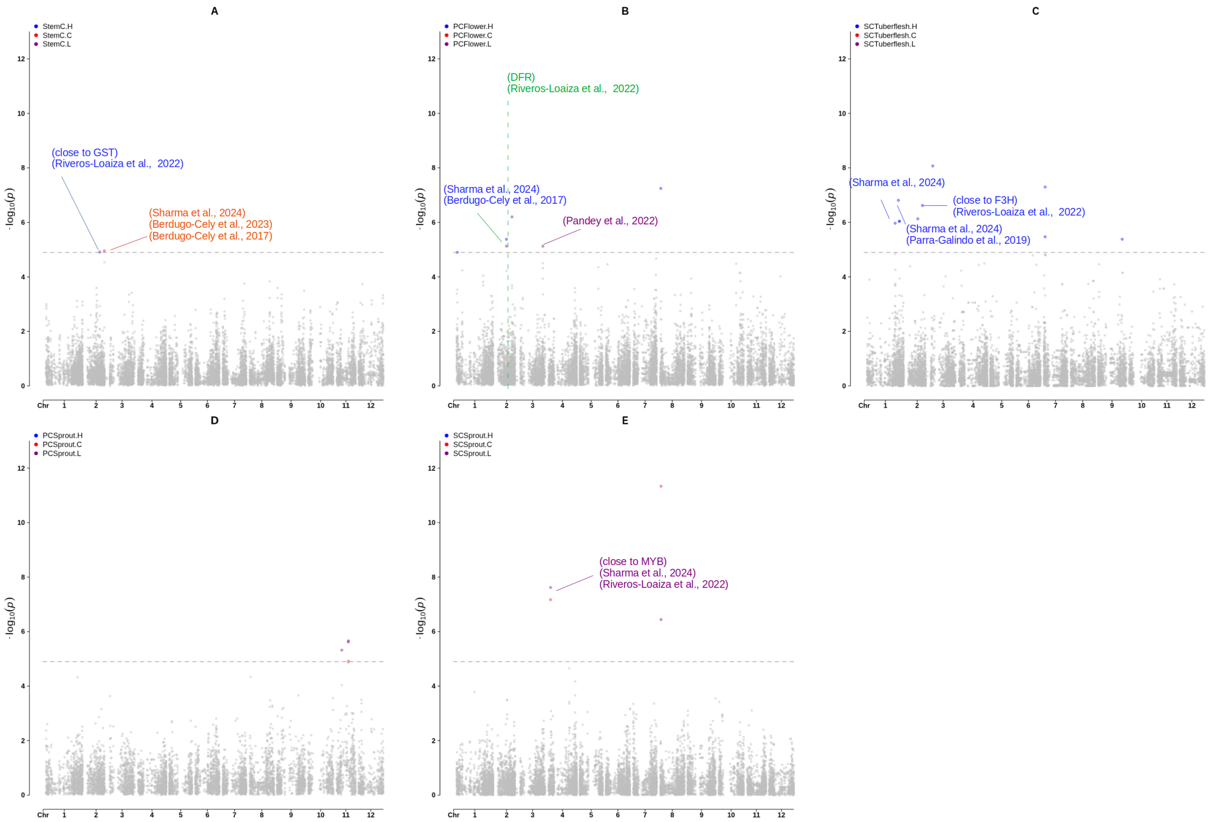

Fig. 4 Manhattan plots of markers associated with LCH traits. Colors indicate significant SNPs for each color component: A. StemC, B. PCFlower, C. SCTuberflesh, D. PCSprout, E. SCSprout. Direct associations previously reported are highlighted in green, while the other references correspond to genes linked to tuber traits located near the identified SNPs, with lines connecting SNPs to their respective references

SNP	Trait (H|C|L)	Tool	Chr	Position	P value	R^2	Gene Id DMv6.1	References
c2_14623	SCTuberflesh (H)	GWASpoly	1	67,822,946	1.07E-06	0.21	DM.01G028040	(Sharma et al., 2024)
c1_4666	SCTuberflesh (H)	GWASpoly	1	75,668,861	1.55E-07	0.23	DM.01G036130	**(∼wd40)** (Liu et al., 2020; Sajeevan et al., 2023)
c1_730	SCTuberflesh (H)	GWASpoly	1	78,044,268	9.12E-07	0.21	DM.01G039330	(Sharma et al., 2024)
c2_2551	SCTuberflesh (H)	GWASpoly	1	78,177,940	9.12E-07	0.21	DM.01G039490	(Sharma et al., 2024)
c1_11495	PCFlower (H,L)	GWASpoly	2	24,792,151	4.17E-06	0.03	DM.02G009890	(Sharma et al., 2024; Berdugo-Cely et al., 2017)
c1_11544	SCTuberflesh (H)	GWASpoly	2	26,621,058	7.41E-07	0.21	DM.02G011880	
c2_27439	StemC (H)	GAPIT	2	34,015,955	1.23E-05	0.03	DM.02G019820	**(∼GST)**(Riveros-Loaiza et al., 2022)
c1_8113	SCTuberflesh (H)	GWASpoly	2	37,793,674	2.40E-07	0.22	DM.02G024500	**(∼F3H)**(Riveros-Loaiza et al., 2022)
c1_7964	PCFlower (L)	GWASpoly	2	38,139,240	6.31E-07	0.04	DM.02G024900	**(DFR)**(Riveros-Loaiza et al., 2022)
c1_11459	StemC (C)	GAPIT	2	45,124,985	1.11E-05	0.03	DM.02G033700	(Sharma et al., 2024; Berdugo-Cely et al., 2023, 2017)
c2_29450	SCTuberflesh (H)	GWASpoly	3	5,231,620	8.51E-09	0.26	DM.03G004830	
c1_8194	PCFlower (L)	GWASpoly, GAPIT	3	55,211,205	7.50E-06	0.03	DM.03G030880	(Pandey et al., 2022)
c2_21934	SCSprout(C,L)	GAPIT	4	5,765,944	2.40E-08	0.05	DM.04G005270	**(∼MYB)**(Sharma et al., 2024; Riveros-Loaiza et al., 2022)
c2_36859	SCTuberflesh (H)	GAPIT	7	3,398,145	3.39E-06	0.19	.DM.07G003030	
c2_36872	SCTuberflesh (H)	GWASpoly, GAPIT	7	3,452,120	5.04E-08	0.24	DM.07G003060	
c2_32702	PCFlower (H)	GWASpoly, GAPIT	8	1,899,939	5.66E-08	0.05	DM.08G001440	
c2_51974	SCSprout (C,L)	GAPIT	8	2,472,934	4.64E-12	0.08	DM.08G001790	
c1_11907	SCTuberflesh (H)	GAPIT	9	58,172,429	4.17E-06	0.19	DM.09G022590	() (Parra-Galindo et al., 2021)
c1_11237	PCSprout (L)	GAPIT	11	13,277,951	4.78E-06	0.03	DM.11G011980	
c1_5936	PCSprout (L)	GWASpoly, GAPIT	11	28,989,434	2.40E-06	0.04	DM.11G015780	
c1_5940	PCSprout (C,L)	GWASpoly, GAPIT	11	29,258,142	2.19E-06	0.04	DM.11G015820	

Table 1 Significant SNPs associated with potato color traits (HCL), including trait, analysis tool, chromosome, genomic position, p-value, proportion of variance explained (R^2), corresponding gene annotations (DM v6.1), and supporting references. SNPs highlighted in yellow are located near previously reported genes associated with tuber traits, while one in green represents a direct association previously reported for color.

Notably, chromosome 2 accounted for 28.5% of the significant SNPs, followed by chromosomes 1 and 11, each contributing 19% and 14.29%, respectively. Two of the chromosomes with the highest number of significant SNPs have previously been reported to harbor loci involved in potato pigmentation, with chromosomes 2 and 11 linked to the synthesis of red and blue anthocyanins (van Eck et al., 1993; Riveros-Loaiza et al., 2022).

One associated SNP was located within the Dihydroflavonol reductase gene DFR (DM.02G024900) at the R locus, known to control the production of red anthocyanins(De Jong et al., 2003; Riveros-Loaiza et al., 2022) (Figure 4 B). In addition, a gene on chromosome 2, located near both DFR and Flavanone-3-hydroxylase (F3H; DM.02G024500), was associated with the secondary color of tuber flesh and may form part of a co-regulatory region. A locus on chromosome 2 near 45 Mb has also been previously associated with tuber skin brightness (Sharma et al., 2024), secondary tuber flesh color (Berdugo-Cely et al., 2017), and with Total Phenol content (Berdugo-Cely et al., 2023). Chromosome 2 thus harbors multiple loci linked to color traits.

On chromosome 1, we detected a locus associated with the secondary color of tuber flesh, previously reported as likely involved in anthocyanin biosynthesis and pigmentation regulation via a WD40-like regulatory protein (DM.01G036130) (Liu et al., 2020; Sajeevan et al., 2023). Furthermore, a SNP on chromosome 3 at 55 Mb corresponded to a major QTL reported for purple skin and tuber flesh color (Pandey et al., 2021; Sharma et al., 2024). Additional loci were found in proximity to QTLs for other tuber-related traits, such as eye depth, average tuber depth, and skin brightness (Sharma et al., 2024; Pandey et al., 2022). These results confirm that the genes involved in coloration in the Andigenum group are consistent with previous potato studies, including loci linked to nutraceutical traits (Berdugo-Cely et al., 2023; Riveros-Loaiza et al., 2022; Parra-Galindo et al., 2019). However, 48% of the SNPs identified have not been reported in earlier studies, highlighting the need to further explore the Andigenum group.

Moreover, GWASpoly and GAPIT identified 14 and 12 significant SNPs, respectively, with six SNPs jointly detected by both tools, representing 28.57% of the total. Although no SNPs were shared across different traits, four SNPs were associated with distinct HCL components within the same trait. A Circos plot (Figure S3) combines the data for the five traits and presents the density of SNP markers for chromosomes, allowing the visualization of SNP co-occurrence across different traits.

3.3 Heritability of Color Traits

Figure 5A summarizes the posterior distributions of narrow-sense heritability (h^2) for 21 LCH components across seven color traits in the Andigenum group of *Solanum tuberosum*. Each boxplot represents the distribution of h^2 values obtained from Bayesian Ridge Regression (BRR), showing the median, interquartile range, and 95% credible interval.

Most primary color traits—particularly `PCFlower.H`, `StemC.L`, and `PCSprout.C`—exhibited moderate to high heritability, with means exceeding 0.55. The highest heritability was observed in the hue component of secondary tuber flesh color (`SCTuberflesh.H`), with a mean near 0.98, indicating strong genetic control.

In contrast, several secondary color traits such as SCSprout.C, SCSprout.H/L, and SCTuberskin.C displayed lower heritability values (posterior means between 0.20 and 0.35), suggesting a greater influence of environmental or non-additive factors.

Overall, StemC, PCFlower, and PCSprout traits consistently showed moderate to strong heritability across their Chroma, Hue, and Lightness components, supporting their potential as targets for genomic selection.

3.4 Genomic Prediction Results

Figure 5B shows the predictive abilities of the seven genomic prediction models for each of the 21 LCH trait components. Predictive ability is defined as the correlation between observed phenotypic values and GEBVs across cross-validation folds.

The highest predictive abilities were observed in traits such as PCFlower.H, StemC.L, and PCSprout.C, with mean correlations above 0.70. Other primary color traits—including PCTuberskin.C and PCSprout.L—showed moderate predictive performance.

Traits with the lowest predictive abilities included several secondary color traits, such as SCSprout.H, SCTuberskin.C, and SCTuberflesh.L, all with mean correlations below 0.40. These results reflect the patterns observed in the heritability estimates (panel A), where traits with higher heritability generally showed greater predictive accuracy. However, the secondary color of the tuber flesh showed a marked deviation, explained by the limited data available for this trait (only 94 accessions characterized) combined with the cross-validation strategy used to estimate predictive ability, which resulted in low predictive values for the three components (< 0.35).

Model performance varied across traits, with BRR, GBLUP, and RKHS performing best for different subsets of traits (Figure 5B).

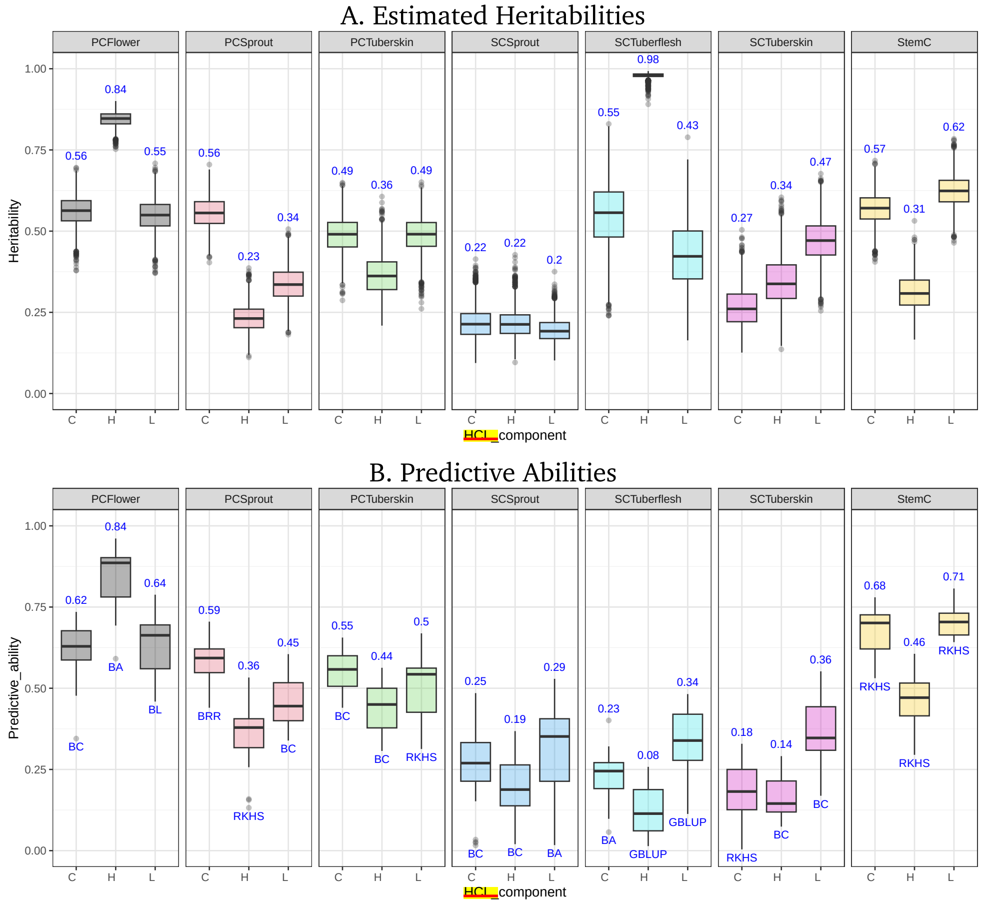

Fig. 5 Heritability and Predictive Ability for LCH Traits. Each boxplot corresponds to one of the 21 LCH color trait components in *Solanum tuberosum* Andigenum group. Traits are grouped by color feature (e.g., flower, sprout, skin, stem) and by component (Chroma: C, Hue: H, Lightness: L). **A. Posterior distributions of narrow-sense heritability (h^2),** estimated using Bayesian Ridge Regression (BRR) with 15,000 iterations, 5,000 burn-in, and thinning every 5 steps. Posterior medians are annotated above each box. **B. Predictive abilities of seven genomic prediction models.** Values represent the correlation between observed phenotypes and GEBVs across repeated 5-fold cross-validation. The mean predictive ability is shown above each boxplot, and the model achieving the best performance per trait is indicated below.

3.4.1 GP using marker subsets

Genomic prediction was performed using marker subsets representing 1%–100% of the 4,641 SNPs. Predictive ability, assessed by the Pearson correlation between observed phenotypes and genomic estimated breeding values (GEBVs), varied by trait and marker density (see Figure S4).

In general, predictive ability increased rapidly with marker density up to 40%-50% of the total set, after which gains plateaued, indicating diminishing returns. This pattern held across most color traits and LCH components (Hue, Chroma, Lightness),

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suggesting that a moderate number of well-distributed, informative SNPs can achieve predictive performance comparable to that of the full marker set.

Stem color and primary flower color consistently showed higher predictive abilities, with correlations approaching 0.75 at full marker density. In contrast, secondary traits with lower data availability (Fig S2) —such as secondary tuber flesh (n=94), secondary tuber skin color (n=473), and secondary sprout color (n=542)—exhibited lower predictive values. Secondary color results may still be influenced by the cross-validation strategy and small sample sizes, even when larger marker sets are used. This variability highlights differences in genetic architecture, with some traits likely governed by major loci and others shaped by more polygenic and complex effects.

These results support the feasibility of cost-effective genomic selection for tetraploid *Andigenum* potatoes by using an optimized SNP subset. The top 50% of markers, selected via the GSCORE metric (Section 2.3), effectively captured the relevant genetic variation, enabling accurate predictions while reducing the computational demand.

4 Discussion

The Andigenum group remains largely unexplored (Ghislain and Douches, 2020), as potato breeding has relied on only a few representatives from this upland adapted group (Spanoghe et al., 2022), leaving much of its extensive diversity untapped. As evidence, numerous potato genomes have been published in recent years (Zhou et al., 2020; Campoy et al., 2020; Sun et al., 2022; Bao et al., 2022), but only a small fraction belong to the Andigenum group (Kanehisa et al., 2016; Reyes-Herrera et al., 2024), even within recent pangenome initiatives (Hoopes et al., 2022; Guo, 2023; Bozan et al., 2023) the percentage of Andigenum accessions representatives is less than 4%. Research on Andigenum potatoes may uncover novel genes, as over 80% of the genes under selection in *Solanum tuberosum* cultivars (Chilotanum) and the Andigenum group are population-specific (Hardigan et al., 2017). The Andigenum thus provides a broad reservoir of alleles for potato breeding. The Andigenum accessions from the CCC have been reported to exhibit high genetic diversity (Berdugo-Cely et al., 2017), underscoring both the role of germplasm banks in preserving traditional cultivars and the potential benefits of investigating conserved materials.

4.1 Testing associated genes with dark pigmentation

The first step toward understanding and promoting the consumption of these types of potatoes is the recognition of their various properties and traits through genetic characterization. In this way, our research team is currently studying the genetic diversity of the Colombian Central Collection of potatoes, with a particular focus on native Andean genotypes that produce tubers with dark pigmentation. At present, we are quantifying traits related to nutritional value, such as zinc, iron, and ascorbic acid contents, to dissect their genetic bases and select genetic materials that can be used as parents in potato breeding programs to develop Colombian biofortified varieties. Additionally, within the same initiative, we aim to validate candidate genes associated with color and nutritional traits identified in our previous research initiatives (Berdugo-Cely et al., 2023, 2017). These genes are currently being validated through biotechnological approaches, including gene expression analyses and genome editing using Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) (Berdugo-Cely, 2025).

The genes identified in the present study also require validation through similar biotechnological strategies. Those confirmed to be associated with tuber color traits could be used to develop molecular markers, which can then be integrated into molecular breeding programs to facilitate the selection of potato genotypes with specific pigmentation patterns.

Finally, it is necessary to continue the genetic and phenotypic analysis of the Colombian Central Collection of potatoes under several approaches, including responses to biotic and abiotic stresses, since this collection constitutes the primary ex situ reservoir of genetic diversity for this species in Colombia.

4.2 How this study contributes to the conservation of threatened potato diversity

The investigation of this study into the genetic architecture of color traits in potatoes is a fundamental contribution to the conservation of native varieties, many of which face accelerated genetic erosion in Colombia and the Andes due to their replacement by commercial varieties and the homogenization of production systems (Arce et al., 2019; Ellis et al., 2018). Identifying SNPs associated with color expression in different organs provides useful markers to strengthen the characterization and monitoring of diversity in ex situ collections. This facilitates the management of germplasm banks and the prioritization of accessions for conservation programs (Ellis et al., 2018). In contrast, dynamic conservation approaches that link germplasm banks with local communities allow ex situ and in situ conservation to be integrated into participatory processes (Lüttringhaus et al., 2021). Similarly, systematic monitoring and field conservation strategies help maintain and utilize the diversity of native potatoes in their territories of origin (Arce et al., 2019; Dawson et al., 2023). Thus, this knowledge integrates genomic tools with community-based approaches, contributing to the recovery, sustainable use, and revaluation of native potatoes in Colombia's Andean territories.

4.3 The opportunities of this study to future projects associated to decolonizing the food consumption and contribute with food security and local suverignty

Studying the genetic diversity of native potatoes will contribute to their conservation and promote cultural revaluation and consumption. Pigmented potato varieties, particularly those with purple or black flesh, have a high content of bioactive compounds such as anthocyanins and carotenoids, which have a positive effect on human health (Burgos et al., 2020; Vollmer et al., 2022). Moreover, dark potatoes have...xxxx. This study corroborates these results by linking the darkness of xxx to xxxxx and xxxx.

Linking genetic knowledge to potato color and nutritional properties adds new insights into how potato varieties represent not only food, but also ancestral knowledge and cultural heritage (Galvis-Tarazona et al., 2022). However, the colonial legacy has significantly impacted the consumption of traditional food systems, such as potatoes, in Colombia and the region. For example, in Peru, it has been found that neocolonial practices have marginalized indigenous cultural knowledge toward Western preferences and marked standards devaluing some varieties because of their appearance, including color.

Despite the nutritional value of dark potato varieties, their consumption is not extended because colonial stereotypes link dark color with low-quality or low-value food (REF).

Linking genetic knowledge to potato color and nutritional properties can overcome cultural barriers and open differentiated markets that recognize the nutritional and cultural value of these varieties. Currently, in Colombia, some high-end restaurants have dark potatoes on their menus (AgroAvances, 2022).

Likewise, native potatoes should be understood as living heritage, reflecting centuries of peasant selection and management practices (Arce et al., 2019; Lüttringhaus et al., 2021). Thus, the challenge is to decolonize food consumption patterns that favor uniform industrial varieties to promote diverse diets that strengthen food sovereignty and resilience of Andean communities (Dawson et al., 2023).

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References

Azizi, A., Abbaspour-Gilandeh, Y., Nooshyar, M., Afkari-Sayah, A.: Identifying potato varieties using machine vision and artificial neural networks. International Journal of Food Properties **19**(3), 618–635 (2016)

Arce, A., Haan, S., Juarez, H., Burra, D.D., Plasencia, F., Ccanto, R., Polreich, S., Scurrah, M.: The spatial-temporal dynamics of potato agrobiodiversity in the highlands of central peru: A case study of smallholder management across farming landscapes. Land **8**(11), 169 (2019)

AgroAvances: Las manos científicas al rescate de las papas nativas de Colombia. AgroAvances (2022). https://agroavances.com/noticias-detalle.php?idNot=4198 Accessed 2025-09-11

Anderson, C.A., Pettersson, F.H., Clarke, G.M., Cardon, L.R., Morris, A.P., Zondervan, K.T.: Data quality control in genetic case-control association studies. Nature Protocols **5**(9), 1564–1573 (2010) https://doi.org/10.1038/nprot.2010.116

Ayala-Silva, T., Meerow, A.W.: Evaluation of flower color using a colorimeter and the royal horticultural society charts. (2006)

Bozan, I., Achakkagari, S.R., Anglin, N.L., Ellis, D., Tai, H.H., Strömvik, M.V.: Pangenome analyses reveal impact of transposable elements and ploidy on the evolution of potato species. Proceedings of the National Academy of Sciences **120**(31), 2211117120 (2023) https://doi.org/10.1073/pnas.2211117120

Berdugo-Cely, J.A.: Proyecto: Descifrando las bases genéticas de la producción de compuestos nutricionales y la respuesta al tizón tardío en papa. Research project funded by the Ministerio de Ciencia y Tecnología de Colombia. Grant awarded to AGROSAVIA in representation of Jhon A. Berdugo-Cely (2024 call) (2025)

Berdugo-Cely, J.A., Céron-Lasso, M.d.S., Yockteng, R.: Phenotypic and molecular analyses in diploid and tetraploid genotypes of solanum tuberosum l. reveal promising genotypes and candidate genes associated with phenolic compounds, ascorbic acid contents, and antioxidant activity. Frontiers in Plant Science **13**, 1007104 (2023)

Bellumori, M., Chasquibol Silva, N.A., Vilca, L., Andrenelli, L., Cecchi, L., Innocenti, M., Balli, D., Mulinacci, N.: A study on the biodiversity of pigmented andean potatoes: Nutritional profile and phenolic composition. Molecules **25**(14), 3169 (2020)

Berdugo-Cely, J., Valbuena, R.I., Sánchez-Betancourt, E., Barrero, L.S., Yockteng, R.: Genetic diversity and association mapping in the Colombian Central Collection of Solanum tuberosum L. Andigenum group using SNPs markers. PLOS ONE **12**(3), 0173039 (2017) https://doi.org/10.1371/journal.pone.0173039

Black, W.: Studies on the inheritance of tuber colour in potatoes. Journal of Genetics **27**, 319–339 (1933)

Bao, Z., Li, C., Li, G., Wang, P., Peng, Z., Cheng, L., Li, H., Zhang, Z., Li, Y., Huang, W., *et al.*: Genome architecture and tetrasomic inheritance of autotetraploid potato. Molecular Plant **15**(7), 1211–1226 (2022)

Bonar, N., Liney, M., Zhang, R., Austin, C., Dessoly, J., Davidson, D., Stephens, J., McDougall, G., Taylor, M., Bryan, G.J., *et al.*: Potato mir828 is associated with purple tuber skin and flesh color. Frontiers in plant science **9**, 1742 (2018)

Behn, A., Lizana, C., Zapata, F., Gonzalez, A., Reyes-Díaz, M., Fuentes, D.: Phenolic and anthocyanin content characterization related to genetic diversity analysis of solanum tuberosum subsp. tuberosum chilotanum group in southern chile. Frontiers in plant science **13**, 1045894 (2023)

Balladares, V., Ramos, M.: Evaluación de antocianinas y micronutrientes en papa nativa (solanum andigena). Alimentos Ciencia e Ingeniería **26**(1), 7–16 (2018)

Bvenura, C., Witbooi, H., Kambizi, L.: Pigmented potatoes: a potential panacea for food and nutrition security and health? Foods **11**(2), 175 (2022)

Burgos, G., Zum Felde, T., Andre, C., Kubow, S.: In: Campos, H., Ortiz, O. (eds.) The Potato and Its Contribution to the Human Diet and Health, pp. 37–74. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-28683-5_2

Cho, K., Cho, K.-S., Sohn, H.-B., Ha, I.J., Hong, S.-Y., Lee, H., Kim, Y.-M., Nam, M.H.: Network analysis of the metabolome and transcriptome reveals novel regulation of potato pigmentation. Journal of experimental botany **67**(5), 1519–1533 (2016)

Caraza-Harter, M.V., Endelman, J.B.: Image-based phenotyping and genetic analysis of potato skin set and color. Crop Science **60**(1), 202–210 (2020)

Consortium, P.G.S.: Genome sequence and analysis of the tuber crop potato. Nature **475**(7355), 189–195 (2011) https://doi.org/10.1038/nature10158

Campoy, J.A., Sun, H., Goel, M., Jiao, W.-B., Folz-Donahue, K., Wang, N., Rubio, M., Liu, C., Kukat, C., Ruiz, D., *et al.*: Gamete binning: chromosome-level and haplotype-resolved genome assembly enabled by high-throughput single-cell sequencing of gamete genomes. Genome biology **21**(1), 1–20 (2020)

Charmet, G., Tran, L.-G., Auzanneau, J., Rincent, R., Bouchet, S.: BWGS: A R package for genomic selection and its application to a wheat breeding programme. PLOS ONE **15**(4), 0222733 (2020) https://doi.org/10.1371/journal.pone.0222733

Devaux, A., Goffart, J.-P., Kromann, P., Andrade-Piedra, J., Polar, V., Hareau, G.: The potato of the future: opportunities and challenges in sustainable agri-food systems. Potato Research **64**(4), 681–720 (2021)

De Jong, W., De Jong, D., De Jong, H., Kalazich, J., Bodis, M.: An allele of dihydroflavonol 4-reductase associated with the ability to produce red anthocyanin pigments in potato (solanum tuberosum l.). Theoretical and Applied Genetics **107**(8), 1375–1383 (2003)

Dawson, T., Juarez, H., Maxted, N., De Haan, S.: Identifying priority sites for the on-farm conservation of landraces and systematic diversity monitoring through an integrated multi-level hotspot analysis: the case of potatoes in peru. Frontiers in Conservation Science **4**, 1130138 (2023)

Dodds, K., Long, D.: The inheritance of colour in diploid potatoes: I. types of anthocyanidins and their genetic loci. Journal of Genetics **53**(1), 136–149 (1955)

Campos, G., Sorensen, D., Gianola, D.: Genomic Heritability: What Is It? PLoS

Genetics **11**(5), 1–21 (2015) https://doi.org/10.1371/journal.pgen.1005048

Ellis, D., Chavez, O., Coombs, J., Soto, J., Gomez, R., Douches, D., Panta, A., Silvestre, R., Anglin, N.L.: Genetic identity in genebanks: application of the solcap 12k snp array in fingerprinting and diversity analysis in the global in trust potato collection. Genome **61**(7), 523–537 (2018)

Felcher, K.J., Coombs, J.J., Massa, A.N., Hansey, C.N., Hamilton, J.P., Veilleux, R.E., Buell, C.R., Douches, D.S.: Integration of two diploid potato linkage maps with the potato genome sequence. PloS one **7**(4), 36347 (2012)

Food and Agriculture Organization of the United Nations: FAOSTAT: Crops and livestock products - Production quantities. https://www.fao.org/faostat/en/#data/QCL. Accessed: 2025-07-29 (2024)

Feldman, M.J., Park, J., Miller, N., Wakholi, C., Greene, K., Abbasi, A., Rippner, D.A., Navarre, D., Schmitz Carley, C., Shannon, L.M., Novy, R.: A scalable, low-cost phenotyping strategy to assess tuber size, shape, and the colorimetric features of tuber skin and flesh in potato breeding populations. The Plant Phenome Journal **7**(1), 20099 (2024) https://doi.org/10.1002/ppj2.20099

Garreta, L., Cerón-Souza, I., Palacio, M.R., Reyes-Herrera, P.H.: MultiGWAS: An integrative tool for Genome Wide Association Studies in tetraploid organisms. Ecology and Evolution **11**, 3–7572 (2021) https://doi.org/10.1002/ece3.7572

Ghislain, M., Douches, D.S.: In: Campos, H., Ortiz, O. (eds.) The Genes and Genomes of the Potato, pp. 139–162. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-28683-5_5

Garnica H, A.M., Romero B, Á.R., Prieto C, L., Cerón L, M.d.S., Argüelles C, J.: Characterization of promising potato clones (solanum tuberosum l. subspecies andigena) for starch extraction. Agronomía Colombiana **30**(3), 326–334 (2012)

Gómez, R.: Guía para las caracterizaciones morfológicas básicas en colecciones de papas nativas. Centro Internacional de la Papa (CIP), Germoplasma de Papa, Dpto. de Mejoramiento y Recursos Genéticos. CIP, Lima, Perú (2000)

Galvis-Tarazona, D.Y., Ojeda-Pérez, Z.Z., Arias-Moreno, D.M.: Cultural and ethnobotanical legacy of native potatoes in colombia. Journal of Ethnobiology and Ethnomedicine **18**(1), 59 (2022)

Guo, Y.: Pangenome and the diversity of potato species. Nature Food **4**(8), 638–638 (2023)

Hardigan, M.A., Laimbeer, F.P.E., Newton, L., Crisovan, E., Hamilton, J.P., Vaillancourt, B., Wiegert-Rininger, K., Wood, J.C., Douches, D.S., Farré, E.M., *et al.*: Genome diversity of tuber-bearing solanum uncovers complex evolutionary history and targets of domestication in the cultivated potato. Proceedings of the National Academy of Sciences **114**(46), 9999–10008 (2017)

Hoopes, G., Meng, X., Hamilton, J.P., Achakkagari, S.R., Guesdes, F.d.A.F., Bolger, M.E., Coombs, J.J., Esselink, D., Kaiser, N.R., Kodde, L., *et al.*: Phased, chromosome-scale genome assemblies of tetraploid potato reveal a complex genome, transcriptome, and predicted proteome landscape underpinning genetic diversity. Molecular plant **15**(3), 520–536 (2022)

Huaman, Z., Williams, J., Salhuana, W., Vincent, L., et al.: Descriptors for the cultivated potato (1977)

Kasajima, I.: Measuring plant colors. Plant Biotechnology **36**(2), 63–75 (2019)

Kanehisa, M., Sato, Y., Morishima, K.: Blastkoala and ghostkoala: Kegg tools for functional characterization of genome and metagenome sequences. Journal of molecular biology **428**(4), 726–731 (2016)

Lachman, J., Hamouz, K.: Red and purple coloured potatoes as a significant antioxidant source in human nutrition. Plant soil enviran **51**(11), 477–482 (2005)

Larico, T.N.B., Huanca, J.M.M., Mancilla, W.M.F., Porres, J.J.A.: Descripción de variedades de papa (solanum tuberosum) en comunidades de la cuenca choquecota del municipio de palca-la paz. CIBUM SCIENTIA **3**(1), 67–80 (2024)

Liu, Z., Liu, Y., Coulter, J.A., Shen, B., Li, Y., Li, C., Cao, Z., Zhang, J.: The wd40 gene family in potato (solanum tuberosum l.): genome-wide analysis and identification of anthocyanin and drought-related wd40s. Agronomy **10**(3), 401 (2020)

Lu, F., Lipka, A.E., Glaubitz, J., Elshire, R.J., Cherney, J.H., Casler, M.D., Buckler, E.S., Costich, D.E.: Switchgrass genomic diversity, ploidy, and evolution: Novel insights from a network-based snp discovery protocol. PLoS Genetics **9**(1), 1003215 (2013) https://doi.org/10.1371/journal.pgen.1003215

Lüttringhaus, S., Pradel, W., Suarez, V., Manrique-Carpintero, N.C., Anglin, N.L., Ellis, D., Hareau, G., Jamora, N., Smale, M., Gómez, R.: Dynamic guardianship of potato landraces by andean communities and the genebank of the international potato center. CABI agriculture and bioscience **2**(1), 45 (2021)

Lyman, O., Ruchard, L., Rexroat, C., Mendenhall, W.: Statistics: A tool for the Social Sciences. PWS-Kent Publishing Company, Boston (379pp) (1986)

Lunden, A.P.: Notes on the inheritance of flower and tuber colour in the potato. Journal of Genetics **25**(3), 339–358 (1932)

Li, M., Xiong, Y., Yang, X., Gao, Y., Li, K.: Transcriptomic and metabolic analysis reveals genes and pathways associated with flesh pigmentation in potato (solanum tuberosum) tubers. Current Issues in Molecular Biology **46**(9), 10335–10350 (2024)

Manrique-Carpintero, N.C., Berdugo-Cely, J.A., Cerón-Souza, I., Lasso-Paredes, Z., Reyes-Herrera, P.H., Yockteng, R.: Defining a diverse core collection of the colombian central collection of potatoes: a tool to advance research and breeding. Frontiers in Plant Science **14**, 1046400 (2023)

Miller, M.D., Schmitz Carley, C.A., Figueroa, R.A., Feldman, M.J., Haagenson, D., Shannon, L.M.: Tubar: An r package for quantifying tuber shape and skin traits from images. American Journal of Potato Research **100**(1), 52–62 (2023)

Nagelkerke, N.J., *et al.*: A note on a general definition of the coefficient of determination. biometrika **78**(3), 691–692 (1991)

Pérez, P., Campos, G.: BGLR : A Statistical Package for Whole Genome Regression and Prediction. Genetics **198**(2), 483–495 (2014)

Pavan, S., Delvento, C., Ricciardi, L., Lotti, C., Ciani, E., D'Agostino, N.: Recommendations for choosing the genotyping method and best practices for quality control in crop genome-wide association studies. Frontiers in Genetics **11**, 447 (2020) https://doi.org/10.3389/fgene.2020.00447

Parra-Galindo, M.-A., Piñeros-Niño, C., Soto-Sedano, J.C., Mosquera-Vasquez, T.: Chromosomes i and x harbor consistent genetic factors associated with the anthocyanin variation in potato. Agronomy **9**(7), 366 (2019)

Parra-Galindo, M.A., Soto-Sedano, J.C., Mosquera-Vásquez, T., Roda, F.: Pathway-based analysis of anthocyanin diversity in diploid potato. PLoS One **16**(4), 0250861 (2021)

Pham, G.M., Hamilton, J.P., Wood, J.C., Burke, J.T., Zhao, H., Vaillancourt, B., Ou, S., Jiang, J., Buell, C.R.: Construction of a chromosome-scale long-read reference genome assembly for potato. GigaScience **9**(9), 100 (2020) https://doi.org/10.1093/gigascience/giaa100

Pandey, J., Scheuring, D.C., Koym, J.W., Coombs, J., Novy, R.G., Thompson, A.L., Holm, D.G., Douches, D.S., Miller Jr, J.C., Vales, M.I.: Genetic diversity and population structure of advanced clones selected over forty years by a potato breeding program in the usa. Scientific Reports **11**(1), 8344 (2021)

Pandey, J., Scheuring, D.C., Koym, J.W., Vales, M.I.: Genomic regions associated with tuber traits in tetraploid potatoes and identification of superior clones for breeding purposes. Frontiers in Plant Science **13**, 952263 (2022)

Rosyara, U.R., De Jong, W.S., Douches, D.S., Endelman, J.B.: Software for Genome-Wide Association Studies in Autopolyploids and Its Application to Potato. The Plant

Genome **9**(2), 1–10 (2016) https://doi.org/10.3835/plantgenome2015.08.0073

Reyes-Herrera, P.H., Delgadillo-Duran, D.A., Flores-Gonzalez, M., Mueller, L.A., Cristancho, M.A., Barrero, L.S.: Chromosome-scale genome assembly and annotation of the tetraploid potato cultivar diacol capiro adapted to the andean region. G3 Genes Genomes Genetics **14**(9), 139 (2024) https://doi.org/10.1093/g3journal/jkae139

Riveros-Loaiza, L.M., Benhur-Cardona, N., Lopez-Kleine, L., Soto-Sedano, J.C., Pinzón, A.M., Mosquera-Vásquez, T., Roda, F.: Uncovering anthocyanin diversity in potato landraces (solanum tuberosum l. phureja) using rna-seq. PLoS One **17**(9), 0273982 (2022)

Ropelewska, E.: Effect of boiling on classification performance of potatoes determined by computer vision. European Food Research and Technology **247**(4), 807–817 (2021)

Royal Horticultural Society: RHS Colour Chart, Sixth edition edn. Royal Horticultural Society, London, UK (2015). Used for standardized color determination in plant phenotyping

Salaman, R.: The inheritance of colour and other characters int the potato. Zeitschrift für induktive Abstammungs-und Vererbungslehre **5**(1), 192–193 (1911)

Sajeevan, R.S., Abdelmeguid, I., Saripella, G.V., Lenman, M., Alexandersson, E.: Comprehensive transcriptome analysis of different potato cultivars provides insight into early blight disease caused by alternaria solani. BMC Plant Biology **23**(1), 130 (2023)

Schmitz Carley, C.A., Coombs, J.J., Douches, D.S., Bethke, P.C., Palta, J.P., Novy, R.G., Endelman, J.B.: Automated tetraploid genotype calling by hierarchical clustering. Theoretical and Applied Genetics **130**(4), 717–726 (2017) https://doi.org/10.1007/s00122-016-2845-5

Sun, H., Jiao, W.-B., Krause, K., Campoy, J.A., Goel, M., Folz-Donahue, K., Kukat, C., Huettel, B., Schneeberger, K.: Chromosome-scale and haplotype-resolved genome assembly of a tetraploid potato cultivar. Nature genetics **54**(3), 342–348 (2022)

Sharma, S.K., McLean, K., Hedley, P.E., Dale, F., Daniels, S., Bryan, G.J.: Genotyping-by-sequencing targets genic regions and improves resolution of genome-wide association studies in autotetraploid potato. Theoretical and Applied Genetics **137**(8), 180 (2024)

Spanoghe, M., Marique, T., Nirsha, A., Esnault, F., Lanterbecq, D.: Genetic diversity trends in the cultivated potato: a spatiotemporal overview. Biology **11**(4), 604 (2022)

Spooner, D.M., McLean, K., Ramsay, G., Waugh, R., Bryan, G.J.: A single domestication for potato based on multilocus amplified fragment length polymorphism genotyping. proceedings of the national academy of sciences **102**(41), 14694–14699 (2005)

Tang, C.: Effects of pigment indicators on the color expression of leaves of the colored-leaf garden plants. Agricultural Science & Technology **15**(11), 1928 (2014)

Thornton, M.K., Lee, J., John, R., Olsen, N.L., Navarre, D.A.: Influence of growth regulators on plant growth, yield, and skin color of specialty potatoes. American journal of potato research **90**(3), 271–283 (2013)

Eck, H.J., Jacobs, J.M., Dijk, J., Stiekema, W.J., Jacobsen, E.: Identification and mapping of three flower colour loci of potato (s. tuberosum l.) by rflp analysis. Theoretical and Applied Genetics **86**(2), 295–300 (1993)

Eck, H.J., Jacobs, J.M., Van Den Berg, P.M., Stiekema, W.J., Jacobsen, E.: The inheritance of anthocyanin pigmentation in potato (solanum tuberosum l.) and mapping of tuber skin colour loci using rflps. Heredity **73**(4), 410–421 (1994)

Voss, D.H.: Relating colorimeter measurement of plant color to the royal horticultural society colour chart. HortScience **27**(12), 1256–1260 (1992)

Voss, D.H.: The royal horticultural society colour chart 2001 (2002)

Vos, P.G., Uitdewilligen, J.G., Voorrips, R.E., Visser, R.G., Eck, H.J.: Development and analysis of a 20k snp array for potato (solanum tuberosum): an insight into the breeding history. Theoretical and Applied Genetics **128**(12), 2387–2401 (2015)

Vollmer, R., Villagaray, R., Castro, M., Cárdenas, J., Pineda, S., Espirilla, J., Anglin, N., Ellis, D., Rennó Azevedo, V.C.: The world's largest potato cryobank at the international potato center (cip)–status quo, protocol improvement through large-scale experiments and long-term viability monitoring. Frontiers in Plant Science **13**, 1059817 (2022)

Wigginton, J.E., Cutler, D.J., Abecasis, G.R.: A note on exact tests of hardy-weinberg equilibrium. The American Journal of Human Genetics **76**(5), 887–893 (2005) https://doi.org/10.1086/429864

Wang, J., Zhang, Z.: Gapit version 3: Boosting power and accuracy for genomic association and prediction. Genomics, Proteomics and Bioinformatics **19**(4), 629–640 (2021) https://doi.org/10.1016/j.gpb.2021.08.005

Yusuf, M., Miller, M.D., Stefaniak, T.R., Haagenson, D., Endelman, J.B., Thompson, A.L., Shannon, L.M.: Genomic prediction for potato (solanum tuberosum) quality traits improved through image analysis. The Plant Genome **17**(4), 20507 (2024)

Yin, L., Zhang, H., Tang, Z., Xu, J., Yin, D., Zhang, Z., Yuan, X., Zhu, M., Zhao, S., Li, X., *et al.*: rmvp: a memory-efficient, visualization-enhanced, and parallel-accelerated tool for genome-wide association study. Genomics, Proteomics and Bioinformatics **19**(4), 619–628 (2021)

Zhang, Y., Jung, C.S., De Jong, W.S.: Genetic analysis of pigmented tuber flesh in potato. Theoretical and Applied Genetics **119**, 143–150 (2009)

Zhou, Q., Tang, D., Huang, W., Yang, Z., Zhang, Y., Hamilton, J.P., Visser, R.G., Bachem, C.W., Buell, C.R., Zhang, Z., *et al.*: Haplotype-resolved genome analyses of a heterozygous diploid potato. Nature genetics **52**(10), 1018–1023 (2020)